

Investigating causal factors of fatal shark attacks

DSAN 5650: Causal Inference for Computational Social Science

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Using the Global Shark Attack File maintained by the Shark Research Institute, this project aims to investigate the causal factors leading to fatal shark attacks. Because the shark attack file only contains data on those who have been attacked, this study will focus on what makes a shark attack lethal. This will be accomplished by constructing a PGM with some of the variables found in the Global Shark Attack File such as location, species, time of day, season, activity, and others. The aim of this project is to identify which factors truly lead to fatal attacks by controlling for each of the factors listed before and considering their interactions to identify a solid model that accurately represents the causal relationships between them.

Introduction

While it is commonly agreed that geography can have important consequences for economic outcomes, there is a growing debate over the channel of causality. Some authors stress a direct contemporaneous channel whereby geography directly affects economic outcomes today. For instance, Gallup and Sachs (2001) and Sachs and Malaney (2002) claim that a disease-prone environment has substantial negative consequences for current income levels because it reduces productivity, lowers the incentives to invest in human capital, draws scarce resources towards medical care, and discourages foreign investment and tourism.¹ Alternatively, other authors stress an indirect channel whereby geography indirectly affects economic outcomes today through its past interactions with key historical events. For instance, Acemoglu, Johnson, and Robinson (2001) argue that the importance of a disease-prone environment for current income levels lies in the effect that it had on potential settler mortality during colonization. In areas where high mortality discouraged Europeans from settling, colonizers implemented

¹Other geographical characteristics that have been linked to economic outcomes through a direct contemporaneous channel include proximity to the coast, and the prevalence of desert or tropical climate (Kamarrck, 1976, Mellinger, Sachs, and Gallup, 2000, Sachs, 2001, Rappaport and Sachs, 2003).

poor institutions which adversely affected subsequent economic development.² Generally, it is difficult to separate the two channels. First, because the physical environment is extremely persistent.³ Second, because the geographical features on which the literature has focused are mostly specific to the areas that were subject to the historical events of interest. For instance, areas having a disease prone-environment today also had it in the past and tropical diseases do not have the potential to affect many areas not subject to colonization.⁴

In this paper we investigate one geographic characteristic, terrain ruggedness, for which we are able to estimate both the direct contemporaneous channel and the indirect historic channel. Rugged terrain has obvious negative direct impacts on current economic outcomes. Irregular terrain makes cultivation difficult. On steep slopes erosion becomes a potential hazard, and the control of water (e.g. irrigation) becomes much more difficult. According to the Food and Agriculture Organization (1993), when slopes are greater than 2 degrees, the benefits of cultivation often do not cover the necessary costs, and when slopes are greater than 6 degrees cultivation becomes impossible. In addition, because of the very high costs involved in earthwork, building costs are much greater when terrain is irregular (Rapaport and Snickars, 1999, Nogales, Archondo-Callao, and Bhandari, 2002). As well, transportation over irregular terrain is slower and more costly.⁵

In Africa, in addition to this negative direct effect, ruggedness may also have had a positive indirect effect on economic outcomes, by allowing areas to escape from the slave trades and their negative consequences. For a period of nearly 500 years, from 1400 to 1900, Africa simultaneously experienced four slave trades (the trans-Atlantic, the trans-Saharan, the Red Sea, and the Indian Ocean slave trades). Over 18 million slaves were exported from Africa over this period, and even more died during raids and transport (total African population was roughly stable over this period at 50–70 million). There is ample historical evidence documenting the adverse long-term effects of Africa’s slave trades.⁶ Nunn (2008) examines the

²Other papers that argue for connections between the physical environment and economic outcomes through historical interactions include Engerman and Sokoloff (1997, 2002), Sokoloff and Engerman (2000), Diamond (1997), and Acemoglu, Johnson, and Robinson (2002).

³An interesting exception is rainfall, which Miguel, Satyanath, and Sergenti (2004) use as an instrumental variable to investigate the effect of economic growth on the likelihood of civil conflict. Rainfall data has also been used to investigate the impact of income on witch killing in Tanzania (Miguel, 2005), of income on property crimes in 19th century Germany (Mehlum, Miguel, and Torvik, 2006a), and income on overseas remittances in the Philippines (Yang, 2006). Other studies focus on the direct link of rainfall on subsequent economic, social, and health outcomes (e.g., Maccini and Yang, 2006, Levine and Yang, 2006).

⁴Because of these difficulties, few studies attempt to quantify or directly test for the relative importance of the two channels. One notable exception is Easterly and Levine (2003), who employ tests of over-identifying restrictions in an instrumental variables framework. They conclude that measures of tropics, germs, and crops do not explain economic development beyond their ability to explain institutional development.

⁵A recent study by Allen, Bourke, and Gibson (2005) highlights these negative effects of irregular terrain within Papua New Guinea. The authors show that steep terrain not only makes the production of cash crops very difficult, but it also makes it much more costly or even impossible to transport the crops to the markets. The end result is that the populations living in these parts of Papua New Guinea have lower incomes and poorer health.

⁶See for example Hawthorne (1999), Inikori (2000), Lovejoy (2000), and Klein (2003).